

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Fundamentals Of Data Science

COURSE CODE: DSA04

COURSE OUTCOME COVERED

CO3: Evaluate datasets using descriptive statistics, including summarization, outlier detection, and distribution analysis, and perform statistical inference through estimation and hypothesis testing. (BL3)

Assessment Tool Weightage: 20%

QUESTIONS: 20

Total Marks:100

Annexure A

Data Interpretation

Code of Conduct:

I HARI VISHNU N / 192524319 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary) 5 Marks	Level 3 (Proficient) 3 /4 Marks	Level 2 (Developing) 2 Marks	Level 1 (Beginning) 1 Mark	Marks
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Annexure A

Problem 01: Hospital Patient Health Monitoring

Scenario

A hospital wants to investigate how patient lifestyle factors affect blood pressure levels.

Dataset

Patient ID	Daily Exercise (Minutes)	BMI	Systolic BP
P001	10	32	150
P002	15	31	148
P003	20	30	145
P004	25	29	142
P005	30	28	138
P006	35	27	134
P007	40	26	130
P008	45	25	126
P009	50	24	122
P010	55	23	118

Tasks

1. Find the mean and standard deviation of blood pressure.
2. Calculate covariance between:
 - o Exercise and Blood Pressure
 - o BMI and Blood Pressure
3. Calculate correlation coefficients.
4. Identify whether exercise or BMI is more strongly associated with blood pressure.
5. Discuss possible healthcare interventions.

CODE:

```
import pandas as pd
import numpy as np
data = pd.read_csv("hospital.csv")
print("\nHospital Patient Dataset")
print(data)
# 1. Mean and Standard Deviation
bp_mean = data["Systolic_BP"].mean()
bp_std = data["Systolic_BP"].std()
print("\n1. Blood Pressure Statistics")
print("Mean Blood Pressure :", round(bp_mean, 2))
```

```

print("Standard Deviation :", round(bp_std, 2))
# 2. Covariance
cov_exercise = data["Daily_Exercise"].cov(data["Systolic_BP"])
cov_bmi = data["BMI"].cov(data["Systolic_BP"])
print("\n2. Covariance")
print("Exercise vs Blood Pressure :", round(cov_exercise, 2))
corr_exercise = data["Daily_Exercise"].corr(data["Systolic_BP"])
corr_bmi = data["BMI"].corr(data["Systolic_BP"])
print("\n3. Correlation Coefficient")
print("Exercise vs Blood Pressure :", round(corr_exercise, 4))

print("BMI vs Blood Pressure      :", round(corr_bmi, 4))

```

OUTPUT:

```
== RESTART: C:\Users\hari vishnu\OneDrive\Desktop\clg\16 F of DS\CO3_AT2\q1.py
```

Hospital Patient Dataset

	Patient_ID	Daily_Exercise	BMI	Systolic_BP
0	P001	10	32	150
1	P002	15	31	148
2	P003	20	30	145
3	P004	25	29	142
4	P005	30	28	138
5	P006	35	27	134
6	P007	40	26	130
7	P008	45	25	126
8	P009	50	24	122
9	P010	55	23	118

1. Blood Pressure Statistics

```

Mean Blood Pressure : 135.3
Standard Deviation  : 11.14

```

2. Covariance

```

Exercise vs Blood Pressure : -168.06
BMI vs Blood Pressure      : 33.61

```

3. Correlation Coefficient

```

Exercise vs Blood Pressure : -0.9969
BMI vs Blood Pressure      : 0.9969

```

Problem 02: E-Commerce Customer Behaviour Analysis

Scenario

An e-commerce company wants to identify factors influencing customer spending.

Dataset

Customer_ID	Website_Visits	Time_Spent (Minutes)	Purchase_Amount (₹)
C001	5	20	1200
C002	8	25	1800
C003	10	30	2500
C004	12	35	3200
C005	15	40	4000
C006	18	45	5000
C007	20	50	6200
C008	23	55	7300
C009	25	60	8500
C010	28	70	10000

Tasks

1. Calculate mean, variance, and standard deviation of Purchase Amount.
2. Find covariance between:
 - Website Visits and Purchase Amount
 - Time Spent and Purchase Amount
3. Calculate correlation coefficients.
4. Determine which customer behavior metric better predicts spending.
5. Suggest marketing strategies based on the findings.

CODE:

```
import pandas as pd
import numpy as np
data = pd.read_csv("ecom.csv")
print("\nE-Commerce Customer Dataset")
print(data)
# 1. Mean, Variance, Standard Deviation
purchase_mean = data["Purchase_Amount"].mean()
purchase_var = data["Purchase_Amount"].var()
purchase_std = data["Purchase_Amount"].std()
print("\n1. Purchase Amount Statistics")
print("Mean :", round(purchase_mean, 2))
print("Variance :", round(purchase_var, 2))
```

```

print("Standard Deviation :", round(purchase_std, 2))
# 2. Covariance
cov_visits = data["Website_Visits"].cov(data["Purchase_Amount"])
cov_time = data["Time_Spent"].cov(data["Purchase_Amount"])
print("\n2. Covariance")
print("Website Visits vs Purchase Amount :", round(cov_visits, 2))
print("Time Spent vs Purchase Amount :", round(cov_time, 2))
# 3. Correlation Coefficient
corr_visits = data["Website_Visits"].corr(data["Purchase_Amount"])
corr_time = data["Time_Spent"].corr(data["Purchase_Amount"])
print("\n3. Correlation Coefficient")
print("Website Visits vs Purchase Amount :", round(corr_visits, 4))
print("Time Spent vs Purchase Amount :", round(corr_time, 4))

```

OUTPUT:

```

== RESTART: C:\Users\hari vishnu\OneDrive\Desktop\clg\16 F of DS\CO3_AT2\q2.p!

```

E-Commerce Customer Dataset

	Customer_ID	Website_Visits	Time_Spent	Purchase_Amount
0	C001	5	20	1200
1	C002	8	25	1800
2	C003	10	30	2500
3	C004	12	35	3200
4	C005	15	40	4000
5	C006	18	45	5000
6	C007	20	50	6200
7	C008	23	55	7300
8	C009	25	60	8500
9	C010	28	70	10000

1. Purchase Amount Statistics

Mean : 4970.0

Variance : 8793444.44

Standard Deviation : 2965.37

2. Covariance

Website Visits vs Purchase Amount : 22568.89

Time Spent vs Purchase Amount : 47266.67

3. Correlation Coefficient

Website Visits vs Purchase Amount : 0.9914

Time Spent vs Purchase Amount : 0.9949

|

Problem 03: Employee Productivity and Performance

Scenario

A software company is investigating the relationship between employee productivity and performance ratings.

Dataset

Employee_ID	Training_Hours	Projects_Completed	Performance_Rating
E001	5	2	60
E002	8	3	65
E003	10	4	70
E004	12	5	74
E005	15	6	79
E006	18	7	84
E007	20	8	88
E008	22	9	91
E009	25	10	95
E010	28	11	98

Tasks

1. Compute the mean, variance, and standard deviation of Performance Ratings.
2. Calculate covariance between:
 - Training Hours and Performance Rating
 - Projects Completed and Performance Rating
3. Compute correlation coefficients.
4. Identify the strongest factor affecting employee performance.
5. Recommend HR policies to improve employee productivity.

CODE:

```
import pandas as pd
import numpy as np
data = pd.read_csv("employee.csv")
print("\nEmployee Productivity Dataset")
print(data)

#1. Mean, Variance, Standard Deviation
mean = data["Performance_Rating"].mean()
variance = data["Performance_Rating"].var()
std = data["Performance_Rating"].std()
```

```

print("\n1. Performance Rating Statistics")
print("Mean :", round(mean,2))
print("Variance :", round(variance,2))
print("Standard Deviation :", round(std,2))
# 2. Covariance
cov_training = data["Training_Hours"].cov(data["Performance_Rating"])
cov_projects = data["Projects_Completed"].cov(data["Performance_Rating"])
print("\n2. Covariance")
print("Training Hours vs Performance :", round(cov_training,2))
print("Projects Completed vs Performance :", round(cov_projects,2))
# 3. Correlation Coefficient
corr_training = data["Training_Hours"].corr(data["Performance_Rating"])
corr_projects = data["Projects_Completed"].corr(data["Performance_Rating"])
print("\n3. Correlation Coefficient")
print("Training Hours vs Performance :", round(corr_training,4))
print("Projects Completed vs Performance :", round(corr_projects,4))

```

OUTPUT:

```

Employee Productivity Dataset
Employee_ID  Training_Hours  Projects_Completed  Performance_Rating
0      E001           5              2              60
1      E002           8              3              65
2      E003          10              4              70
3      E004          12              5              74
4      E005          15              6              79
5      E006          18              7              84
6      E007          20              8              88
7      E008          22              9              91
8      E009          25             10              95
9      E010          28             11              98

1. Performance Rating Statistics
Mean : 80.4
Variance : 167.82
Standard Deviation : 12.95

2. Covariance
Training Hours vs Performance : 97.87
Projects Completed vs Performance : 39.11

3. Correlation Coefficient
Training Hours vs Performance : 0.9957
Projects Completed vs Performance : 0.9972

```

Problem 04 : Smart City Traffic Analysis

A city administration wants to analyze traffic congestion patterns.

Day	Traffic_Volume (Vehicles)	Rainfall (mm)	Average_Speed (km/h)	Accident_Count
D1	12000	0	55	2
D2	13500	5	50	3
D3	14500	10	48	4
D4	16000	15	45	5
D5	17500	20	42	6
D6	19000	25	38	8
D7	21000	30	35	10
D8	22500	35	32	12
D9	24000	40	28	14
D10	26000	45	25	16

Tasks

1. Calculate descriptive statistics (Mean, Variance, Standard Deviation) for all variables.
2. Compute covariance matrices.
3. Compute correlation matrices.
4. Identify positive and negative relationships among variables.
5. Determine which factor is most associated with accident occurrence.
6. Provide data-driven recommendations to reduce traffic accidents.
7. Visualize relationships using scatter plots and interpret the results.

CODE:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Read CSV file
data = pd.read_csv("traffic.csv")
print("\nSmart City Traffic Dataset")
print(data)
#1.Descriptive Statisti
print("\n1. Descriptive Statistics")
columns = ["Traffic_Volume", "Rainfall", "Average_Speed", "Accident_Count"]
for col in columns:
    print("\n", col)
    print("Mean :", round(data[col].mean(),2))
    print("Variance :", round(data[col].var(),2))
```

```

    print("Standard Deviation :", round(data[col].std(),2))
# 2. Covariance Matrix
print("\n2. Covariance Matrix")
cov_matrix = data[columns].cov()
print(cov_matrix)
# 3. Correlation Matrix
print("\n3. Correlation Matrix")
corr_matrix = data[columns].corr()
print(corr_matrix)
#Positive & Negative Relationships
print("\n4. Relationships")
print("Traffic Volume and Accident Count :",
      round(corr_matrix.loc["Traffic_Volume","Accident_Count"],4))
print("Rainfall and Accident Count :",
      round(corr_matrix.loc["Rainfall","Accident_Count"],4))
print("Average Speed and Accident Count :",
      round(corr_matrix.loc["Average_Speed","Accident_Count"],4))
print("\nPositive Correlation -> Value close to +1")
print("Negative Correlation -> Value close to -1")
#Scatter Plots
plt.figure(figsize=(5,4))
plt.scatter(data["Traffic_Volume"], data["Accident_Count"])
plt.title("Traffic Volume vs Accident Count")
plt.xlabel("Traffic Volume")
plt.ylabel("Accident Count")
plt.grid(True)
plt.show()
plt.figure(figsize=(5,4))
plt.scatter(data["Rainfall"], data["Accident_Count"])
plt.title("Rainfall vs Accident Count")
plt.xlabel("Rainfall")
plt.ylabel("Accident Count")
plt.grid(True)
plt.show()

```

```
plt.figure(figsize=(5,4))
plt.scatter(data["Average_Speed"], data["Accident_Count"])
plt.title("Average Speed vs Accident Count")
plt.xlabel("Average Speed")
plt.ylabel("Accident Count")
plt.grid(True)
plt.show()
```

OUTPUT:

Smart City Traffic Dataset

	Day	Traffic_Volume	Rainfall	Average_Speed	Accident_Count
0	D1	12000	0	55	2
1	D2	13500	5	50	3
2	D3	14500	10	48	4
3	D4	16000	15	45	5
4	D5	17500	20	42	6
5	D6	19000	25	38	8
6	D7	21000	30	35	10
7	D8	22500	35	32	12
8	D9	24000	40	28	14
9	D10	26000	45	25	16

1. Descriptive Statistics

Traffic_Volume

Mean : 18600.0

Variance : 22155555.56

Standard Deviation : 4706.97

Rainfall

Mean : 22.5

Variance : 229.17

Standard Deviation : 15.14

Average_Speed

Mean : 39.8

Variance : 97.73

Standard Deviation : 9.89

Accident_Count

Mean : 8.0

Variance : 23.33

Standard Deviation : 4.83

2. Covariance Matrix

	Traffic_Volume	Rainfall	Average_Speed	Accident_Count
Traffic_Volume	2.215556e+07	71111.111111	-46422.222222	22611.111111
Rainfall	7.111111e+04	229.166667	-149.444444	72.222222
Average_Speed	-4.642222e+04	-149.444444	97.733333	-47.222222
Accident_Count	2.261111e+04	72.222222	-47.222222	23.333333

3. Correlation Matrix

	Traffic_Volume	Rainfall	Average_Speed	Accident_Count
Traffic_Volume	1.000000	0.997977	-0.997616	0.994471
Rainfall	0.997977	1.000000	-0.998580	0.987658
Average_Speed	-0.997616	-0.998580	1.000000	-0.988864
Accident_Count	0.994471	0.987658	-0.988864	1.000000

4. Relationships

Traffic Volume and Accident Count : 0.9945

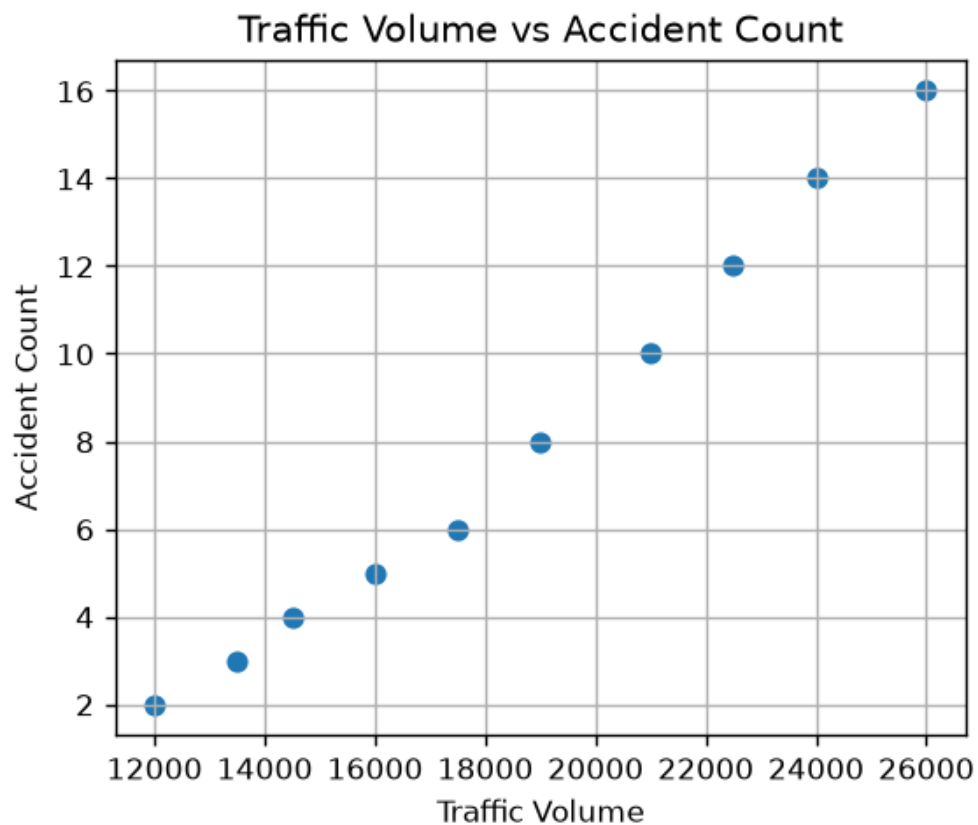
Rainfall and Accident Count : 0.9877

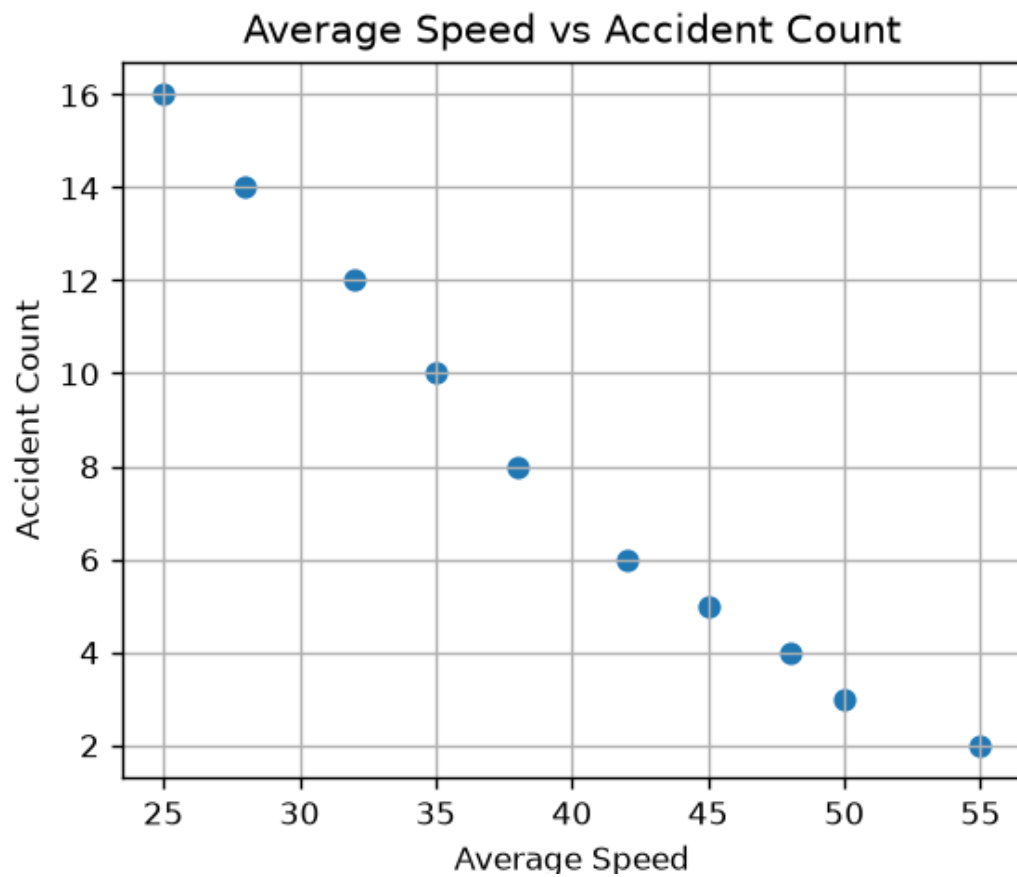
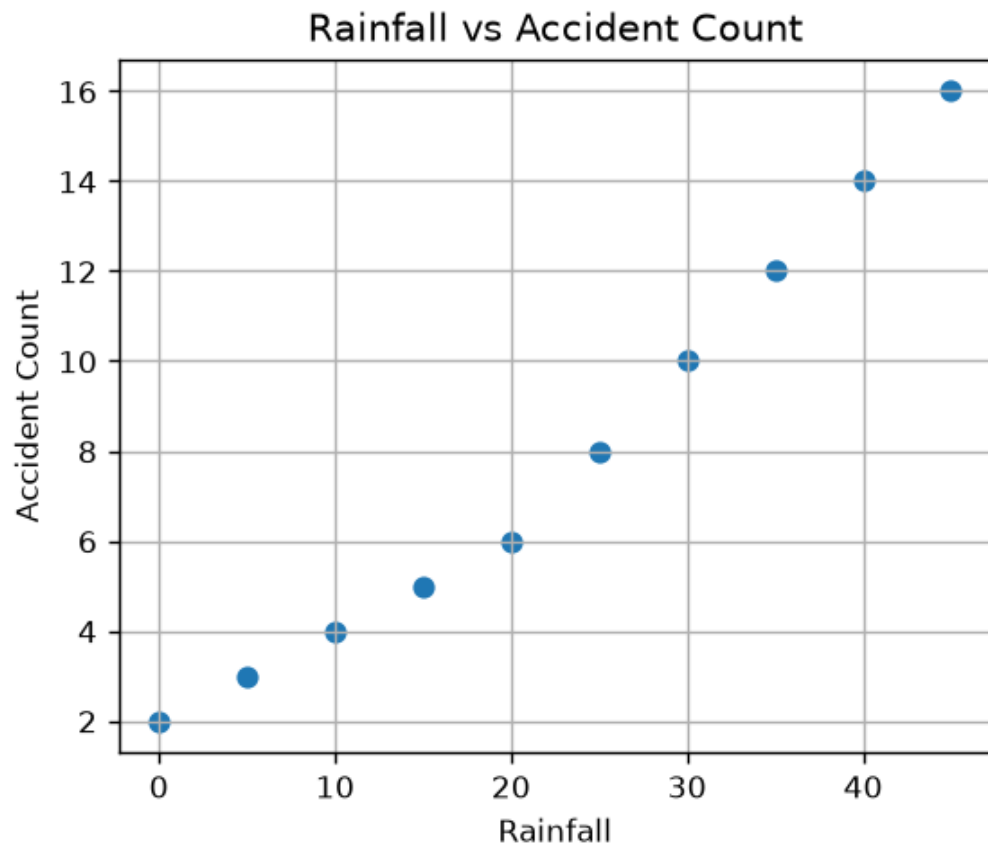
Average Speed and Accident Count : -0.9889

Positive Correlation -> Value close to +1

Negative Correlation -> Value close to -1

|





Problem 5: Average Delivery Time

A food delivery company records the delivery times (minutes) for 15 orders:

22, 25, 18, 30, 27, 24, 21, 19, 26, 23, 28, 20, 24, 25, 22

Tasks:

1. Find the point estimate of the population mean delivery time.
2. Find the point estimate of the population variance.
3. Interpret the estimates.

CODE:

```
import pandas as pd
import numpy as np
data = pd.read_csv("delivery.csv")
print("\nDelivery Time Dataset")
print(data)
mean = data["Delivery_Time"].mean()
variance = data["Delivery_Time"].var(ddof=1)
print("\n1. Point Estimate of Population Mean")
print("Mean Delivery Time :", round(mean, 2), "minutes")
print("\n2. Point Estimate of Population Variance")
print("Variance :", round(variance, 2))
```

OUTPUT:

```
Delivery Time Dataset
   Delivery_Time
0              22
1              25
2              18
3              30
4              27
5              24
6              21
7              19
8              26
9              23
10             28
11             20
12             24
13             25
14             22

1. Point Estimate of Population Mean
Mean Delivery Time : 23.6 minutes

2. Point Estimate of Population Variance
Variance : 11.4
```

Problem 6: Customer Spending

A sample of 12 customers spent the following amounts (\$):

120, 145, 130, 155, 160, 125, 140, 135, 150, 170, 165, 155

Tasks:

1. Estimate the average spending.
2. Estimate the standard deviation.
3. Explain why these are point estimates.

CODE:

```
import pandas as pd
import numpy as np
data = pd.read_csv("spending.csv")
print("\nCustomer Spending Dataset")
print(data)
mean = data["Customer_Spending"].mean()
std = data["Customer_Spending"].std()
print("\n1. Estimated Average Spending")
print("Average Spending : $", round(mean, 2))
print("\n2. Estimated Standard Deviation")
print("Standard Deviation :", round(std, 2))
```

OUTPUT:

```
Customer Spending Dataset
   Customer_Spending
0                120
1                145
2                130
3                155
4                160
5                125
6                140
7                135
8                150
9                170
10               165
11               155

1. Estimated Average Spending
Average Spending : $ 145.83

2. Estimated Standard Deviation
Standard Deviation : 16.07
```

Problem 7: Product Ratings

A company collects ratings from 20 customers:

4, 5, 3, 4, 5, 4, 3, 5, 4, 4, 5, 3, 4, 5, 4, 3, 5, 4, 5, 4

Tasks:

1. Compute the sample mean.
2. Compute the sample variance.
3. Compute the standard error of the mean.
4. Explain what the standard error indicates.

CODE:

```
import pandas as pd
import numpy as np
data = pd.read_csv("ratings.csv")
print("\nProduct Ratings Dataset")
print(data)
# 1. Sample Mean
mean = data["Rating"].mean()
# 2. Sample Variance
variance = data["Rating"].var()
# 3. Standard Error
std = data["Rating"].std()
n = len(data)
standard_error = std / np.sqrt(n)
# Display Results
print("\n1. Sample Mean")
print("Mean Rating :", round(mean, 2))
print("\n2. Sample Variance")
print("Variance :", round(variance, 4))
print("\n3. Standard Error of the Mean")
print("Standard Error :", round(standard_error, 4))
```

OUTPUT:

Product Ratings Dataset

Rating

0	4
1	5
2	3
3	4
4	5
5	4
6	3
7	5
8	4
9	4
10	5
11	3
12	4
13	5
14	4
15	3
16	5
17	4
18	5
19	4

1. Sample Mean

Mean Rating : 4.15

2. Sample Variance

Variance : 0.5553

3. Standard Error of the Mean

Standard Error : 0.1666

|

Problem 8: Exam Scores

Scores of 25 students:

72, 75, 68, 80, 77, 73, 79, 81, 76, 74, 78, 69, 70, 82, 71, 75, 77, 73, 80, 76, 74, 79, 72, 81, 78

Tasks:

1. Calculate mean and standard deviation.
2. Calculate standard error.
3. Compare standard deviation and standard error.

CODE:

```

import pandas as pd
import numpy as np
data = pd.read_csv("scores.csv")
print("\nExam Scores Dataset")
print(data)
mean = data["Score"].mean()
std = data["Score"].std()
n = len(data)
standard_error = std / np.sqrt(n)
print("\n1. Mean and Standard Deviation")
print("Mean :", round(mean, 2))
print("Standard Deviation :", round(std, 2))
print("\n2. Standard Error")
print("Standard Error :", round(standard_error, 2))

```

OUTPUT:

```

Exam Scores Dataset
  Score
0     72
1     75
2     68
3     80
4     77
5     73
6     79
7     81
8     76
9     74
10    78
11    69
12    70
13    82
14    71
15    75
16    77
17    73
18    80
19    76
20    74
21    79
22    72
23    81
24    78

1. Mean and Standard Deviation
Mean : 75.6
Standard Deviation : 3.96

2. Standard Error
Standard Error : 0.79

```

Problem 9: Manufacturing Weights

A sample of 50 packets has:

- Mean weight = 498 g
- Standard deviation = 12 g

Tasks:

1. Construct a 95% confidence interval for the population mean weight.
2. Interpret the interval.
3. Determine whether the target weight of 500 g is plausible.

CODE:

```
import pandas as pd
import numpy as np
from scipy.stats import norm
data = pd.read_csv("weights.csv")

n = data.loc[0, "Sample_Size"]
mean = data.loc[0, "Mean_Weight"]
std = data.loc[0, "Standard_Deviation"]
z = norm.ppf(0.975)
margin = z * (std / np.sqrt(n))
lower = mean - margin
upper = mean + margin

print("\nManufacturing Weights")
print("\nSample Mean :", mean, "g")
print("Sample Standard Deviation :", std, "g")
print("Sample Size :", n)

print("\n1. 95% Confidence Interval")
print("(", round(lower,2), ", ", round(upper,2), ")")
print("\n2. Interpretation")
print("We are 95% confident that the true population mean weight")
print("lies between", round(lower,2), "g and", round(upper,2), "g.")
```

OUTPUT:

Manufacturing Weights

Sample Mean : 498 g

Sample Standard Deviation : 12 g

Sample Size : 50

1. 95% Confidence Interval

(494.67 , 501.33)

2. Interpretation

We are 95% confident that the true population mean weight lies between 494.67 g and 501.33 g.

Problem 10: Customer Satisfaction

Two branches report:

Branch Mean Rating 95% CI

A	4.3	(4.1, 4.5)
B	4.0	(3.8, 4.2)

Tasks:

1. Compare the confidence intervals.
2. Can you conclude Branch A performs better?
3. Explain your reasoning.

CODE:

```
import pandas as pd
data = pd.read_csv("satisfaction.csv")
print("\nCustomer Satisfaction Data")
print(data)
mean_A = data.loc[0, "Mean_Rating"]
lower_A = data.loc[0, "Lower_CI"]
upper_A = data.loc[0, "Upper_CI"]
mean_B = data.loc[1, "Mean_Rating"]
lower_B = data.loc[1, "Lower_CI"]
upper_B = data.loc[1, "Upper_CI"]
print("\n1. Confidence Intervals")
```

```

print(f"Branch A : ({lower_A}, {upper_A})")
print(f"Branch B : ({lower_B}, {upper_B})")
print("\n2. Does Branch A Perform Better?")
if lower_A > upper_B:
    print("Yes. Branch A performs significantly better.")
elif upper_A < lower_B:
    print("No. Branch B performs significantly better.")
else:
    print("The confidence intervals overlap.")
    print("We cannot conclude that Branch A performs better.")

```

OUTPUT:

```

Customer Satisfaction Data
  Branch  Mean_Rating  Lower_CI  Upper_CI
0      A           4.3         4.1         4.5
1      B           4.0         3.8         4.2

1. Confidence Intervals
Branch A : (4.1, 4.5)
Branch B : (3.8, 4.2)

2. Does Branch A Perform Better?
The confidence intervals overlap.
We cannot conclude that Branch A performs better.
|

```

Problem 11: Average Battery Life

A battery manufacturer claims the average battery life is 10 hours.

A sample of 36 batteries gives:

- Mean = 9.5 hours
- Standard deviation = 1.8 hours

Tasks:

1. Formulate null and alternative hypotheses.
2. Conduct a hypothesis test at $\alpha = 0.05$.

3. State your conclusion.

CODE:

```
import pandas as pd
import numpy as np
from scipy.stats import t

# Read CSV file
data = pd.read_csv("battery.csv")
n = data.loc[0, "Sample_Size"]
mean = data.loc[0, "Mean_Battery_Life"]
std = data.loc[0, "Standard_Deviation"]
mu = data.loc[0, "Claimed_Mean"]

# 1. Hypotheses
print("\n1. Hypotheses")
print("H0:  $\mu = 10$  hours")
print("H1:  $\mu \neq 10$  hours")

# 2. One-Sample t-Test
t_stat = (mean - mu) / (std / np.sqrt(n))

# Two-tailed p-value
p_value = 2 * (1 - t.cdf(abs(t_stat), df=n-1))

print("\n2. Hypothesis Test")
print("t-Statistic :", round(t_stat, 4))
print("p-value :", round(p_value, 4))

alpha = 0.05

print("\n3. Conclusion")

if p_value < alpha:
```

```
print("Reject the Null Hypothesis (H0).")
print("The average battery life is significantly different from 10 hours.")
else:
    print("Fail to Reject the Null Hypothesis (H0).")
    print("There is insufficient evidence to conclude that the average battery life differs from 10 hours.")
```

OUTPUT:

1. Hypotheses

$H_0: \mu = 10$ hours

$H_1: \mu \neq 10$ hours

2. Hypothesis Test

t-Statistic : -1.6667

p-value : 0.1045

3. Conclusion

Fail to Reject the Null Hypothesis (H_0).

There is insufficient evidence to conclude that the average battery life differs from 10 hours.

Problem 12: Marketing Campaign

A company believes a new campaign increases sales.

Sales before campaign:

50, 52, 49, 55, 53, 51, 54, 50

Sales after campaign:

58, 60, 57, 62, 59, 61, 63, 58

Tasks:

1. Conduct a hypothesis test.
2. Calculate the p-value.
3. Interpret the p-value.
4. Make a decision at $\alpha = 0.05$.

CODE:

```
import pandas as pd
from scipy.stats import ttest_rel

# Read CSV file
data = pd.read_csv("campaign.csv")

print("\nMarketing Campaign Dataset")
print(data)

# Get Before and After Sales
before = data["Before"]
after = data["After"]

print("\n1. Hypotheses")
print("H0: There is no increase in sales after the campaign.")
print("H1: Sales increased after the campaign.")

# 2. Paired t-Test
t_stat, p_value = ttest_rel(after, before)

print("\n2. Paired t-Test")
print("t-Statistic :", round(t_stat, 4))
print("p-value :", round(p_value, 4))
```

OUTPUT:

Marketing Campaign Dataset

	Before	After
0	50	58
1	52	60
2	49	57
3	55	62
4	53	59
5	51	61
6	54	63
7	50	58

1. Hypotheses

H0: There is no increase in sales after the campaign.

H1: Sales increased after the campaign.

2. Paired t-Test

t-Statistic : 18.9315

p-value : 0.0

Problem 13: Call Centre Response Time

A sample of 100 calls has:

- Mean response time = 4.8 minutes
- Standard deviation = 1.5 minutes

Tasks:

1. Estimate the population mean.
2. Construct a 95% confidence interval.
3. Explain the Frequentist interpretation.

CODE:

```
import pandas as pd
```

```
import numpy as np
```

```
from scipy.stats import norm
```

```
data = pd.read_csv("response.csv")
```

```
n = data.loc[0, "Sample_Size"]
```

```
mean = data.loc[0, "Mean_Response_Time"]
```

```
std = data.loc[0, "Standard_Deviation"]
```

```

print("\nCall Centre Response Time")

print(data)

print("\n1. Estimated Population Mean")

print("Population Mean =", mean, "minutes")

z = norm.ppf(0.975)

margin = z * (std / np.sqrt(n))

lower = mean - margin

upper = mean + margin

print("\n2. 95% Confidence Interval")

print("Lower Limit :", round(lower, 3))

print("Upper Limit :", round(upper, 3))

```

OUTPUT:

Call Centre Response Time

	Sample_Size	Mean_Response_Time	Standard_Deviation
0	100	4.8	1.5

1. Estimated Population Mean

Population Mean = 4.8 minutes

2. 95% Confidence Interval

Lower Limit : 4.506

Upper Limit : 5.094

Problem 14: Defective Products

A sample of 500 products contains 22 defective items.

Tasks:

1. Estimate the defect rate.
2. Construct a 95% confidence interval.
3. Explain the Frequentist meaning of the interval.

CODE

```
import pandas as pd
import numpy as np
from scipy.stats import norm

data = pd.read_csv("defective.csv")
n = data.loc[0, "Sample_Size"]
defective = data.loc[0, "Defective_Items"]
p = defective / n

print("\nDefective Products Analysis")
print("Sample Size :", n)
print("Defective Items :", defective)
print("\n1. Estimated Defect Rate")

print("Defect Rate =", round(p, 4), "(", round(p*100, 2), "% )")
z = norm.ppf(0.975)
margin = z * np.sqrt((p * (1 - p)) / n)
lower = p - margin
upper = p + margin

print("\n2. 95% Confidence Interval")
print("Lower Limit =", round(lower, 4))
print("Upper Limit =", round(upper, 4))
```

OUTPUT:

Defective Products Analysis
Sample Size : 500
Defective Items : 22

1. Estimated Defect Rate
Defect Rate = 0.044 (4.4 %)

2. 95% Confidence Interval
Lower Limit = 0.026
Upper Limit = 0.062

|

Problem 15: Email Spam Detection

Historical information:

- Probability an email is spam = 0.25
- Probability spam email contains keyword "Offer" = 0.80
- Probability non-spam email contains keyword "Offer" = 0.10

Tasks:

1. Compute the probability that an email is spam given it contains "Offer."
2. Apply Bayes' theorem.
3. Interpret the result.

CODE:

```
import pandas as pd
data = pd.read_csv("spam.csv")
P_spam = data.loc[0, "P_Spam"]
P_offer_given_spam = data.loc[0, "P_Offer_Given_Spam"]
P_offer_given_nonspam = data.loc[0, "P_Offer_Given_NonSpam"]
P_nonspam = 1 - P_spam
P_offer = (P_offer_given_spam * P_spam) + \
          (P_offer_given_nonspam * P_nonspam)
P_spam_given_offer = (P_offer_given_spam * P_spam) / P_offer
print("\nEmail Spam Detection")
```

```

print("\nGiven:")
print("P(Spam) =", P_spam)
print("P(Offer | Spam) =", P_offer_given_spam)
print("P(Offer | Non-Spam) =", P_offer_given_nospam)
print("\n1. Probability that an email is Spam given it contains 'Offer'")
print("P(Spam | Offer) =", round(P_spam_given_offer, 4))
print("\n2. Bayes' Theorem")
print("P(Spam | Offer) = [P(Offer | Spam) × P(Spam)] / P(Offer)")
print("\n3. Interpretation")
print("If an email contains the word 'Offer',")
print("there is a", round(P_spam_given_offer * 100, 2),
      "% chance that it is spam.")

```

OUTPUT:

Email Spam Detection

Given:

$P(\text{Spam}) = 0.25$

$P(\text{Offer} | \text{Spam}) = 0.8$

$P(\text{Offer} | \text{Non-Spam}) = 0.1$

1. Probability that an email is Spam given it contains 'Offer'

$P(\text{Spam} | \text{Offer}) = 0.7273$

2. Bayes' Theorem

$P(\text{Spam} | \text{Offer}) = [P(\text{Offer} | \text{Spam}) \times P(\text{Spam})] / P(\text{Offer})$

3. Interpretation

If an email contains the word 'Offer',
there is a 72.73 % chance that it is spam.

Problem 16: Disease Diagnosis

Given:

- Disease prevalence = 2%
- Test sensitivity = 95%
- Test specificity = 90%

Tasks:

1. Compute the posterior probability of disease after a positive test.
2. Explain why the result differs from 95%.

CODE:

```
import pandas as pd
data = pd.read_csv("disease.csv")

prevalence = data.loc[0, "Disease_Prevalence"]
sensitivity = data.loc[0, "Sensitivity"]
specificity = data.loc[0, "Specificity"]
P_no_disease = 1 - prevalence
false_positive = 1 - specificity
P_positive = (sensitivity * prevalence) + (false_positive * P_no_disease)
posterior = (sensitivity * prevalence) / P_positive

print("\nDisease Diagnosis")
print("\nGiven:")
print("Disease Prevalence :", prevalence)
print("Test Sensitivity :", sensitivity)
print("Test Specificity :", specificity)
print("\n1. Posterior Probability")
print("P(Disease | Positive Test) =", round(posterior, 4))

print("\n2. Percentage")
print("Probability =", round(posterior * 100, 2), "%")
print("\n3. Explanation")
print("Although the test has 95% sensitivity,")
print("the disease is very rare (only 2% prevalence).")

print("Many positive results come from healthy people")
print("because of false positives.")
print("Therefore, the probability of actually having")
print("the disease after a positive test is much lower than 95%.")
```

OUTPUT:

Disease Diagnosis

Given:

Disease Prevalence : 0.02

Test Sensitivity : 0.95

Test Specificity : 0.9

1. Posterior Probability

$P(\text{Disease} \mid \text{Positive Test}) = 0.1624$

2. Percentage

Probability = 16.24 %

3. Explanation

Although the test has 95% sensitivity, the disease is very rare (only 2% prevalence). Many positive results come from healthy people because of false positives.

Therefore, the probability of actually having the disease after a positive test is much lower than 95%.

Problem 17: Website Conversion Rate

A company tests two landing pages.

Group Visitors Purchases

A	5000	250
---	------	-----

B	5200	312
---	------	-----

Tasks:

1. Calculate conversion rates.
2. Formulate hypotheses.
3. Conduct a two-proportion z-test.
4. Determine whether page B performs better.

CODE:

```
import pandas as pd
```

```
import numpy as np
```

```
from scipy.stats import norm
```

```
data = pd.read_csv("conversion.csv")
```

```

n1 = data.loc[0, "Visitors"]
x1 = data.loc[0, "Purchases"]
n2 = data.loc[1, "Visitors"]
x2 = data.loc[1, "Purchases"]

p1 = x1 / n1
p2 = x2 / n2

print("\nWebsite Conversion Rate Analysis")
print("\n1. Conversion Rates")
print("Page A :", round(p1 * 100, 2), "%")
print("Page B :", round(p2 * 100, 2), "%")
print("\n2. Hypotheses")
print("H0: pA = pB")
print("H1: pB > pA")

p_pool = (x1 + x2) / (n1 + n2)
SE = np.sqrt(p_pool * (1 - p_pool) * ((1 / n1) + (1 / n2)))
z = (p2 - p1) / SE
p_value = 1 - norm.cdf(z)

print("\n3. Two-Proportion Z-Test")
print("Z-Statistic :", round(z, 4))
print("P-Value :", round(p_value, 6))
alpha = 0.05

print("\n4. Decision")
if p_value < alpha:
    print("Reject the Null Hypothesis.")
    print("Page B performs significantly better than Page A.")
else:
    print("Fail to Reject the Null Hypothesis.")
    print("There is no significant difference between the two pages.")

```

OUTPUT:

Website Conversion Rate Analysis

1. Conversion Rates

Page A : 5.0 %

Page B : 6.0 %

2. Hypotheses

$H_0: p_A = p_B$

$H_1: p_B > p_A$

3. Two-Proportion Z-Test

Z-Statistic : 2.2127

P-Value : 0.013459

4. Decision

Reject the Null Hypothesis.

Page B performs significantly better than Page A.

Problem 18: Mobile App Feature Test

Feature A:

- 400 users
- Average engagement = 18 min
- SD = 5 min

Feature B:

- 420 users
- Average engagement = 20 min
- SD = 6 min

Tasks:

1. Test whether Feature B increases engagement.
2. Calculate test statistic and p-value.
3. Interpret results.

CODE:

```
import pandas as pd
import numpy as np
from scipy.stats import norm
```

```

# Read CSV file
data = pd.read_csv("engagement.csv")

# Read values
n1 = data.loc[0, "Users"]
mean1 = data.loc[0, "Average_Engagement"]
sd1 = data.loc[0, "Standard_Deviation"]

n2 = data.loc[1, "Users"]
mean2 = data.loc[1, "Average_Engagement"]
sd2 = data.loc[1, "Standard_Deviation"]

print("\nMobile App Feature Test")

print("\n1. Hypotheses")
print("H0:  $\mu_A = \mu_B$ ")
print("H1:  $\mu_B > \mu_A$ ")

SE = np.sqrt((sd1**2 / n1) + (sd2**2 / n2))
z = (mean2 - mean1) / SE
# One-tailed p-value
p_value = 1 - norm.cdf(z)

print("\n2. Test Results")
print("Mean Engagement (Feature A):", mean1, "minutes")
print("Mean Engagement (Feature B):", mean2, "minutes")
print("Z-Statistic :", round(z, 4))
print("P-Value :", round(p_value, 6))
alpha = 0.05
print("\n3. Decision")

```

```
if p_value < alpha:
    print("Reject the Null Hypothesis.")
    print("Feature B significantly increases user engagement.")
else:
    print("Fail to Reject the Null Hypothesis.")
    print("Feature B does not significantly increase user engagement.")
```

OUTPUT:

```
Mobile App Feature Test

1. Hypotheses
H0:  $\mu_A = \mu_B$ 
H1:  $\mu_B > \mu_A$ 

2. Test Results
Mean Engagement (Feature A): 18 minutes
Mean Engagement (Feature B): 20 minutes
Z-Statistic : 5.195
P-Value : 0.0

3. Decision
Reject the Null Hypothesis.
Feature B significantly increases user engagement.
```

Problem 19: Recommendation System

A new recommendation algorithm increases click-through rate from 4.8% to 5.3%.

Sample size:

- Old system: 100,000 users
- New system: 100,000 users

Tasks:

1. Test statistical significance.
2. Calculate p-value.
3. Discuss practical significance versus statistical significance.

CODE:

```
import pandas as pd
import numpy as np
from scipy.stats import norm

# Read CSV file
data = pd.read_csv("recommendation.csv")

# Read values
n1 = data.loc[0, "Users"]
ctr1 = data.loc[0, "Click_Through_Rate"] / 100

n2 = data.loc[1, "Users"]
ctr2 = data.loc[1, "Click_Through_Rate"] / 100

# Number of clicks
x1 = int(n1 * ctr1)
x2 = int(n2 * ctr2)

print("\nRecommendation System Analysis")

print("\n1. Hypotheses")
print("H0: CTR_old = CTR_new")
print("H1: CTR_new > CTR_old")

p_pool = (x1 + x2) / (n1 + n2)

SE = np.sqrt(p_pool * (1 - p_pool) * ((1 / n1) + (1 / n2)))

z = (ctr2 - ctr1) / SE

# One-tailed p-value
p_value = 1 - norm.cdf(z)
```

```
print("\n2. Test Results")
print("Old CTR :", round(ctr1 * 100, 2), "%")
print("New CTR :", round(ctr2 * 100, 2), "%")
print("Z-Statistic :", round(z, 4))
print("P-Value :", round(p_value, 8))
alpha = 0.05

print("\n3. Statistical Significance")

if p_value < alpha:
    print("Reject the Null Hypothesis.")
    print("The increase in CTR is statistically significant.")
else:
    print("Fail to Reject the Null Hypothesis.")
    print("The increase in CTR is not statistically significant.")

increase = (ctr2 - ctr1) * 100

print("\n4. Practical Significance")
print("CTR increased by", round(increase, 2), "percentage points.")

if increase >= 1:
    print("The improvement is both statistically and practically significant.")
else:
    print("The improvement is statistically significant but practically small.")
```

CODE:**Recommendation System Analysis****1. Hypotheses**

$H_0: \text{CTR_old} = \text{CTR_new}$

$H_1: \text{CTR_new} > \text{CTR_old}$

2. Test Results

Old CTR : 4.8 %

New CTR : 5.3 %

Z-Statistic : 5.1058

P-Value : 1.6e-07

3. Statistical Significance

Reject the Null Hypothesis.

The increase in CTR is statistically significant.

4. Practical Significance

CTR increased by 0.5 percentage points.

The improvement is statistically significant but practically small.

Problem 20: Salary Prediction Model

Two machine learning models produce Mean Absolute Errors (MAE):

Model MAE

Model A 4200

Model B 4000

Across 50 test folds, the difference in MAE has:

- Mean difference = 200
- Standard deviation = 350

Tasks:

1. Perform a paired t-test.
2. Determine whether Model B significantly outperforms Model A.
3. Explain the result in terms of statistical significance.

CODE:

```
import pandas as pd
```

```
import numpy as np
```

```
from scipy.stats import t
```

```
data = pd.read_csv("model.csv")
```

```
mean_diff = data.loc[0, "Mean_Difference"]
```

```
std = data.loc[0, "Standard_Deviation"]
```

```
n = data.loc[0, "Test_Folds"]
```

```
print("\nSalary Prediction Model Comparison")
```

```
print("\nModel A MAE :", 4200)
```

```

print("Model B MAE :", 4000)
print("\n1. Hypotheses")
print("H0: Mean Difference = 0")
print("H1: Mean Difference > 0 (Model B is better)")
SE = std / np.sqrt(n)
t_stat = mean_diff / SE
# One-tailed p-value
p_value = 1 - t.cdf(t_stat, df=n-1)
print("\n2. Paired t-Test")
print("t-Statistic :", round(t_stat, 4))
print("P-Value :", round(p_value, 8))
alpha = 0.05
print("\n3. Decision")
if p_value < alpha:
    print("Reject the Null Hypothesis.")
    print("Model B significantly outperforms Model A.")
else:
    print("Fail to Reject the Null Hypothesis.")
    print("There is no significant difference between the models.")

```

OUTPUT:

Salary Prediction Model Comparison

Model A MAE : 4200

Model B MAE : 4000

1. Hypotheses

H0: Mean Difference = 0

H1: Mean Difference > 0 (Model B is better)

2. Paired t-Test

t-Statistic : 4.0406

P-Value : 9.378e-05

3. Decision

Reject the Null Hypothesis.

Model B significantly outperforms Model A.

|

HAND WRITTEN PROBLEM SOLUTION

1) Mean and standard deviation of blood pressure.

27/07/20
CO3-AT02

Mean:

$$\text{Mean} = \frac{\sum x}{n}$$

$$\begin{aligned} \text{Sum of value: } & 150 + 148 + 145 + 142 + 139 + 134 + \\ & 130 + 126 + 122 + 118 \\ & = 1353 \end{aligned}$$

$$\text{Mean} = \frac{1353}{10} = 135.3$$

Standard deviation:

$$S = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$$

$$S \approx 11.25$$

Covariance:

$$\text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

a) exercise vs blood pressure:

Mean exercise

covariance:

$$\bar{x} = 132.5$$

$$-183.064$$

b) BMI vs blood pressure:

Mean BMI

$$\bar{x} = 27.5$$

Covariance:

$$26.61$$

correlation coefficient:

$$r = \frac{\text{cov}(x, y)}{s_x s_y}$$

$$r \approx -0.998$$

BMI vs. Blood pressure:

$$r \approx +0.998$$

Problem 02: E-commerce customer Behaviour Analysis.

i) Mean = $\frac{\sum x}{n}$

$$= 1200 + 1800 + 2500 + 3200 + 4000 + 5000 + 6200 + 7500 + 8500 + 10000$$

$$= \frac{49700}{10} = 4970$$

Variance: Sum of squares = 79,141,000

$$s^2 = \frac{79,141,000}{10 - 1}$$

$$= \frac{79,141,000}{9} = 8,793,444.44$$

Std deviation = $s = \sqrt{\text{variance}}$

$$= \sqrt{8,793,444.44}$$

$$= 2965.37$$

ii) Covariance:

$$\text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

$$\bar{x} = \frac{5+8+10+12+15+18+20+23+25+28}{10}$$

$$\bar{x} = \frac{164}{10} = 16.4 //$$

$$\text{Sum of Prod} = \frac{203,120}{9}$$

$$= 22,568.89 //$$

Time spent purchase amount;

$$\bar{x} = \frac{20+25+30+35+40+45+50+55+60+70}{10}$$

$$\bar{x} = \frac{430}{10} = 43 //$$

$$\text{Sum} = \frac{4,25,400}{9} = 47,266.67 //$$

PP) correlation coefficient:

$$r = \frac{\text{Cov}(x, y)}{S_x S_y} \Rightarrow r = 0.998 //$$

Since $0.999 > 0.998$, time spent on the website stronger predictor customer spending.

3) Employee productivity and performance

Task 1: Compute mean of performance rating:

$$(i) \text{Mean}(\bar{x}) = 30.4$$

$$\bar{x} = \frac{60+65+70+79+84+88+91+98+98}{10}$$

$$= \frac{304}{10} = 30.4$$

$$\text{Variance} = 172.29$$

$$\text{Standard deviation} \approx 13.12$$

(2) Covariance

Training hours of performance rating = 136.22

Projects completed & performance rating = 54.22

3) Correlation Coefficients:

Training hours & performance = 0.999

Projects completed & performance = 0.999

4) Strangers Factor Affecting Employee performance

Both variable have an almost perfect positive relationship with performance.

Training hours (0.9999) is slightly stronger than

(4) Smart City Traffic Analysis:-

$$\bar{x} = \frac{\sum x}{n}$$

$$\text{Mean} = \frac{12000 + 13500 + 14500 + 16000 + 17500 + 19000 + 21000 + 22500 + 24000 + 26000}{10}$$

$$= \frac{186000}{10} = 18,600 \text{ vehicles}$$

$$\text{Variance, } \sigma^2 = \frac{\sum (x - \bar{x})^2}{n} = \frac{199,400,000}{10} = 19,940,000$$

$$\text{Std Deviation, } \sigma = \sqrt{\text{variance}}$$

$$= \sqrt{19,940,000}$$

$$= 4,465.42 \text{ vehicles}$$

18) Bam Fall:

$$\text{Mean} = \frac{225}{10} = 22.5 \text{ mm}$$

$$\text{Variance} = \sum (x - \bar{x})^2 = 2002.5 = \frac{26062.5}{10} = 2606.25$$

$$\text{Std. Deviation} = \sqrt{2606.25} = 51.05 \text{ mm}$$

$$\text{Average speed} = \frac{1398}{10} = 139.8 \text{ km/h}$$

$$\text{Variance} = \sum (x - \bar{x})^2 = 879.60 = \frac{879.60}{10} = 87.96$$

$$\text{Standard deviation} = \sqrt{87.96} = 9.38 \text{ km/h}$$

Accident Count:

$$\text{Mean} = 80/10 = 8$$

$$\text{Variance} = \frac{210}{10} = 21$$

$$\text{Std. Deviation} = \sqrt{21} = 4.58$$

12) Covariance Matrix

$$\text{Cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n} = 640000$$

$$\text{cov}(x, y) = \frac{640000}{10} = 64000$$

$$\text{Traffic, volume} = \sum = -417800$$

$$= \frac{-417800}{10} = -41780$$

Traffic volume, accident count

$$\sum = 203800$$

$$\text{cov}(t, A) = \frac{203800}{10} = 20380$$

Ramfall, Average speed, $\Sigma = -134.5$

$$\text{COV}(R, S) = \frac{-134.5}{10} = -13.45$$

Ramfall, Accident count

$$\Sigma = 680$$

$$\text{COV}(R, A) = \frac{680}{10} = 68$$

Covariance (Average speed, Accident count)

Problem 5:

22, 25, 28, 30, 27, 24, 21, 14, 26, 23, 28, 20, 24, 25, 32

$$n = 15$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{354}{15} = 23.6 \text{ min}$$

Sample variance: $\Sigma (x - \bar{x})^2 = 184.60$

$$s^2 = \frac{184.60}{15-1} = \frac{184.60}{14} = 13.19$$

Estimated population variation = 11.4 mm^2

Problem 6:

120, 145, 130, 155, 160, 120, 140, 135, 150, 170, 165, 155

$$\text{Mean, } \bar{x} = \frac{\Sigma x}{n}$$

$$\begin{aligned} \Sigma x &= 120 + 145 + 130 + 155 + 160 + 120 + 140 + 135 + 150 + 170 + 165 + 155 \\ &= 1750 \end{aligned}$$

$$\bar{x} = \frac{1750}{12} = 145.83$$

Standard Deviation,

$$s^2 = \frac{\sum (x - \bar{x})^2}{n-1} = \frac{2839.67}{19} = 149.45$$

$$s = \sqrt{149.45} = 12.22$$

Problem 7: Product Ratings

Given data :- 4, 5, 8, 4, 5, 4, 3, 5, 4, 4, 5, 3, 4, 5, 4, 3, 5, 4, 5, 4

$$n = 20$$

Sample mean :- $\bar{x} = \frac{\sum x}{n}$

$$\sum x = 82$$

$$\bar{x} = \frac{82}{20} = 4.1$$

Sample variance = $s^2 = \frac{\sum (x - \bar{x})^2}{n-1}$

$$s^2 \approx 0.73$$

Standard error :-

$$SE = \frac{s}{\sqrt{n}}$$

$$s = \sqrt{0.73} \approx 0.85$$

$$SE = \frac{0.85}{\sqrt{20}} = \frac{0.85}{4.47} \approx 0.19$$

Problem 8: Exam scores

Given data :-

72, 75, 68, 80, 77, 73, 79, 81, 76, 74, 78,

69, 70, 82, 71, 76, 77, 73, 80, 76, 74, 79, 81, 78.

$$[n = 25]$$

1) Calculate mean: $\bar{x} = \frac{\sum x}{n}$

$$\sum x = 1880$$

$$\bar{x} = \frac{1880}{45} = 41.78$$

2) Std. Deviation: $s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}}$

$$s \approx 4.14$$

3) Std. error: $SE = \frac{s}{\sqrt{n}}$

$$SE = \frac{4.14}{5} = 0.83$$

SD measures data spread; SE measures mean accuracy

Problem 9:-

Given data:-

$$n = 50$$

$$\bar{x} = 498.9$$

$$s = 12.9$$

1) Construct a 95% confidence interval.

$$CI = \bar{x} \pm z \left(\frac{s}{\sqrt{n}} \right)$$

$$z = 1.96$$

$$\text{Std. error: } SE = \frac{12}{\sqrt{50}} = \frac{12}{7.07} = 1.70$$

$$\text{Margin error: } 1.96 \times 1.70 = 3.33$$

$$\text{Confidence level } 498 \pm 3.33$$

$$\text{Lower limit } 498 - 3.33 = 494.67$$

$$\text{Upper limit } 498 + 3.33 = 501.33$$

Problem 10: Customer Satisfaction:-

Branch	Mean Rating	95% confidence interval
A	4.3	(4.1, 4.5)
B	4.0	(3.8, 4.2)

Yes, Branch A has a higher mean rating (4.3) than branch (4.0)

⇒ Although the confidence intervals overlap slightly a bit, branch A has slightly higher average rating.

Problem 11: Average Battery Life

Given data:-

claimed mean (μ) = 10 hours

Sample mean ($\bar{x}$) = 9.5 hours

SD (s) = 1.8 hours

sample size (n) = 36

Significance level (α) = 0.05

1) Hypothesis :-

$$H_0 = \mu = 10$$

$$H_1 : \mu \neq 10$$

2) Test statistic

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}} = \frac{9.5 - 10}{1.8/\sqrt{36}} = \frac{-0.5}{0.3} = -1.67$$

Critical value : ± 1.96

$$|t| = 1.67 < 1.96$$

Fail to reject H_0 .

Problem 12:- Marketing Campaign

Sales before :- 50, 52, 49, 55, 53, 51, 54, 50

Sales after :- 58, 60, 57, 62, 59, 61, 63, 58

(1) $H_0 : \mu \text{ before} = \mu \text{ after}$

$H_1 : \mu \text{ after} > \mu \text{ before}$

(2) $p\text{-value} < 0.001$

(3) The very small p -value indicates that the increase in sales is statistically significant.

(4) Decision :

$$P < 0.05$$

Problem 13:- Call Centre Response Time

sample size = 100

mean = 4.8 min

S.D = 1.5 min

(1) population mean = $\mu \approx 4.8 \text{ min}$

(2) 95% confidence interval :

$$CI = \bar{x} \pm 1.96 \left(\frac{s}{\sqrt{n}} \right)$$

$$S.E. = \frac{1.5}{10} = 0.15$$

$$\text{Margin of error} = 1.96 \times 0.15 = 0.294$$

$$\text{Confidence Level} = 4.8 \pm 0.294$$

$$\text{Lower limit} = 4.506$$

$$\text{Upper limit} = 5.094$$

$$95\% \text{ CI} = (4.51, 5.09) \text{ minutes}$$

Problem 14:- Defective products

Given data:-

$$\text{Sample size} = 500$$

$$\text{Defective products} = 22$$

$$(1) \text{ Defect rate } \hat{p} = \frac{x}{n}$$

$$\hat{p} = \frac{22}{500} = 0.044$$

$$(2) 95\% \text{ Confidence interval:-}$$

$$\hat{p} \pm 1.96 \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$S.E. = \sqrt{\frac{0.044(0.956)}{500}} = 0.00914$$

$$\text{Margin of Error} = 1.96 \times 0.00914 = 0.01797$$

$$\text{Confidence Interval} = 0.044 \pm 0.01797 = 0.026$$

$$\text{Upper limit} = 0.044 + 0.01797 = 0.062$$

$$95\% \text{ CI} = (2.6\%, 6.2\%)$$

Problem 15: E-mail Spam Detection:

Given data:

$$P(\text{spam}) = 0.25$$

$$P(\text{offer} / \text{spam}) = 0.80$$

$$P(\text{offer} / \text{not spam}) = 0.10$$

$$(i) P(\text{spam} / \text{offer}) = \frac{P(\text{offer} / \text{spam}) \cdot P(\text{spam})}{P(\text{offer})}$$

$$P(\text{offer}) = (0.80 \times 0.25) + (0.10 \times 0.75)$$

$$= 0.20 + 0.075$$

$$= 0.275$$

Posterior probability:-

$$P(\text{spam} / \text{offer}) = \frac{0.80 \times 0.25}{0.275}$$

$$= \frac{0.20}{0.275} = \boxed{0.727}$$

Problem 16 - Disease Diagnosis:-

Given data:-

$$P(D) = 0.02$$

$$P(+|D) = 0.95$$

$$P(-|D) = 0.90$$

$$P(+|\bar{D}) = 1 - 0.90 = 0.10$$

(1) Posterior Probability :-

Baye's theorem :-

$$P(D|+) = \frac{P(+|D) P(D)}{P(+)}$$

$$P(+)=0.95(0.02)+0.10(0.98)$$

$$=0.019+0.098=0.117$$

$$P(D|+)=\frac{0.019}{0.117}=0.1624$$

Posterior probability = 16.24%

Problem-17 : Website Conversion Rate :-

Group	Visitors	purchases
A	5000	250
B	5200	312

(1) Conversion Rates :-

$$\text{Group A : } \frac{250}{5000} = 0.05 = 5\%$$

$$\text{Group B : } \frac{312}{5200} = 0.06 = 6\%$$

(2) Hypotheses :-

$$H_0 : P_A = P_B$$

$$H_1 : P_B > P_A$$

(3) Z - Test :-

$$p = \frac{250 + 312}{5000 + 5200} = \frac{562}{10200} = 0.0551$$

(4) Test statistic:

$$Z \approx 2.15$$

p-value

$$p \approx 0.016$$

Problem 18 - Mobile App Feature Test:-

Given data:-

Feature	Users	mean
A	400	18 min
B	420	20 min

(1) Hypotheses:-

$$H_0: \mu_A = \mu_B$$

$$H_1: \mu_B = \mu_A$$

(2) Test statistic:-

$$Z = \frac{20 - 18}{\sqrt{\frac{53}{400} + \frac{6^2}{420}}}$$

$$Z \approx 5.13$$

$$p\text{-value} < 0.001$$

$$p < 0.05$$

Reject H_0

Problem 19- Recommendation System :-

Given data:-

System	CTR	users
old	4.8%	1000000
new	5.3%	10

(1) Statistical Significance:-

$$\begin{aligned}\text{Difference} &= 5.3\% - 4.8\% \\ &= 0.5\%\end{aligned}$$

Test Statistic:-

$$Z \approx 5.06$$

$$p\text{-value} < 0.001$$

Reject H_0

Problem 20- Salary Prediction Model:-

Given data:-

model	MAE
A	4200
B	4000

$$\text{mean difference} = 200, \quad SD = 350, \quad n = 50$$

(1) Paired Test, $H_0: \mu_d = 0$, $H_1: \mu_d > 0$

$$\text{Test Statistic:- } t = \frac{200}{350/\sqrt{50}} = \frac{200}{49.50} \Rightarrow \boxed{t \approx 4.04}$$

$$p\text{-value} < 0.001$$

Reject H_0 .

model B significantly outperforms.